

Assignment Day-1

1. **List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.**



Three real-life scenarios where a multimeter is used:

1. **Computer power supply issue:** Check the PSU output voltage to confirm whether it is supplying the correct voltage to the motherboard.
2. **Laptop charging problem:** Measure the charger/adaptor voltage to check whether it is delivering the required power to the laptop.
3. **Smartphone not charging:** Test the charging port and battery voltage to identify whether the problem is with the port, battery, or charging circuit.

2. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

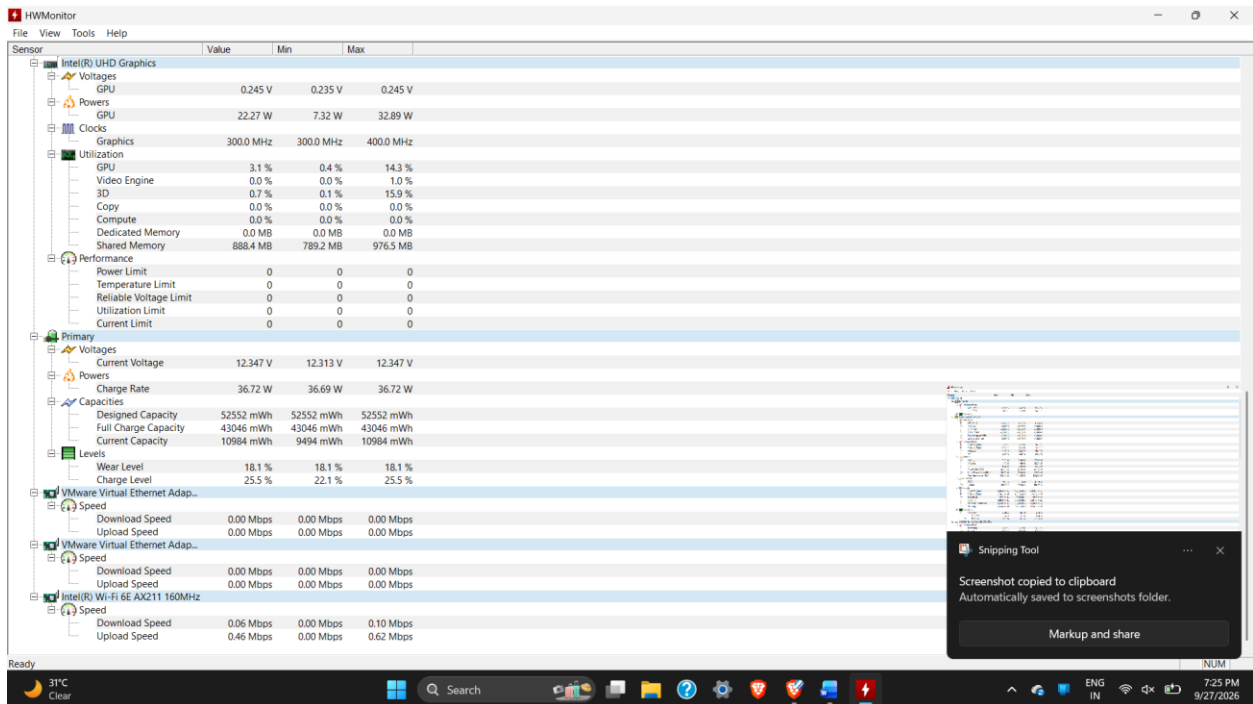
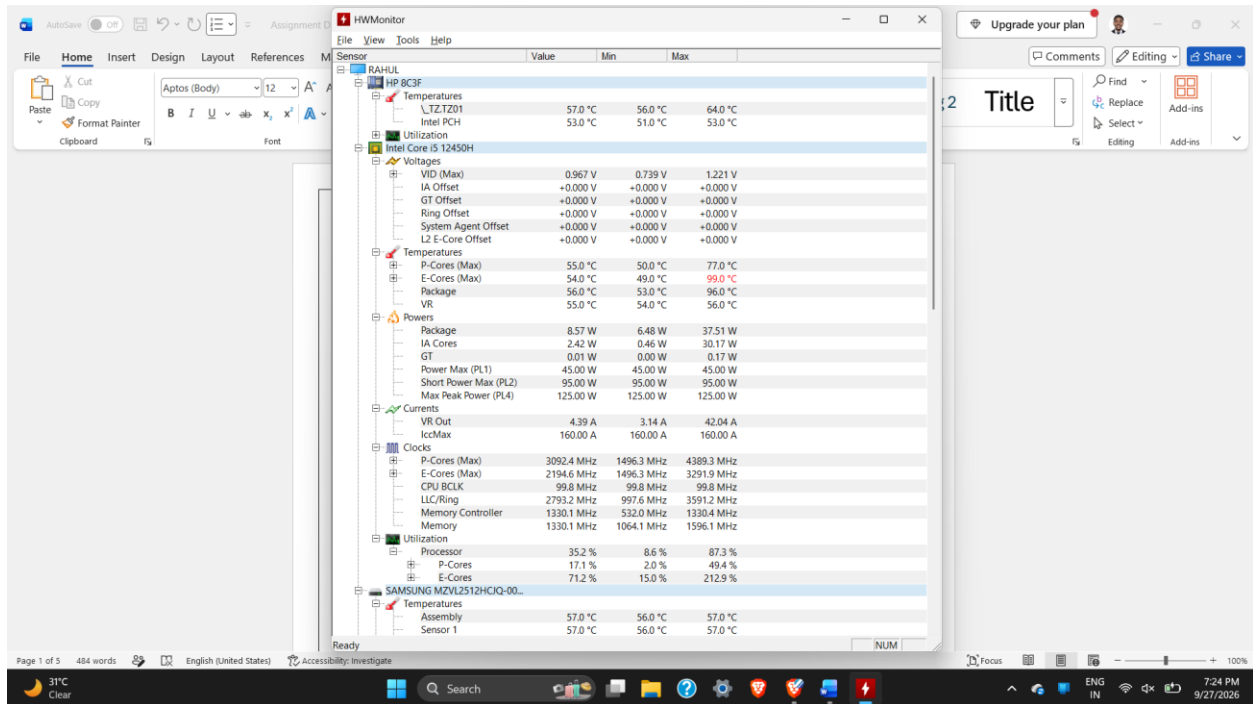


A POST card displays hexadecimal codes while the computer's BIOS checks different hardware components during startup.

In the demonstration, the displayed code changed as the BIOS tested the motherboard hardware, and **8080** appeared when the POST process completed successfully.

The POST card also showed LED indicators for power voltages, clock, reset, and other signals, helping identify hardware or power-related problems.

3. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details. Constraint: Blur or crop out any personal information before submitting.



4. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.



Tool	What it checks	Example Problem Detected
Multimeter	Checks voltage, resistance, and continuity in electronic components.	Faulty power supply or damaged charging cable.
POST Card	Checks motherboard startup/POST process using diagnostic codes.	RAM or motherboard failure during boot.
Software Diagnostics	Checks system health, temperatures, storage, and hardware status through software.	Overheating CPU/GPU or failing hard drive/SSD.